

Technical Report TR-91

Graph Neural Network Topology-Aware Protection:
3-Layer GraphSAGE for Relay Node Classification

SAMBP Research Programme — Phase 8: Advanced Extensions

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1 Introduction and Motivation

Modern power systems are increasingly meshed, reconfigurable, and inverter-dominated. Protection schemes based on fixed thresholds or hand-crafted impedance zones fail when:

- Network topology changes (line outages, tie-switches, meshing);
- Fault current magnitude varies with inverter current limiting;
- Zero-sequence paths are altered by transformer earthing changes.

This report presents **TR-91**: a Graph Neural Network (GNN) relay classifier that is *topology-aware* and *inductively generalisable* — trained on small benchmark networks and deployed, without retraining, on arbitrarily large or previously unseen topologies.

1.1 Objectives

1. Implement 3-layer GraphSAGE [1] for relay node classification with three classes: **RESTRAIN** (0), **ALARM** (1), **TRIP** (2).
2. Train on IEEE 14-bus and IEEE 39-bus (New England) fault scenarios.
3. Demonstrate **zero-shot generalisation** to the IEEE 118-bus system — never presented during training.
4. Validate inference latency $< 100 \mu\text{s}/\text{node}$ for real-time use.

2 GraphSAGE Architecture

2.1 Inductive Representation Learning

GraphSAGE (Hamilton et al., 2017) learns an *inductive* node embedding by aggregating neighbourhood features rather than memorising fixed node indices. This makes the model graph-size agnostic: a model trained on $N = 14$ buses can directly classify nodes in an $N = 118$ bus system.

Each GraphSAGE layer ℓ applies:

$$\mathbf{h}_{\mathcal{N}(i)}^{(\ell)} = \text{MEAN}\left\{\mathbf{h}_j^{(\ell-1)} : j \in \mathcal{N}(i)\right\} = \hat{A} \mathbf{H}_i^{(\ell-1)} \quad (1)$$

$$\mathbf{z}_i^{(\ell)} = \left[\mathbf{h}_i^{(\ell-1)} \parallel \mathbf{h}_{\mathcal{N}(i)}^{(\ell)}\right] \mathbf{W}^{(\ell)} + \mathbf{b}^{(\ell)} \quad (2)$$

$$\mathbf{h}_i^{(\ell)} = \text{ReLU}\left(\mathbf{z}_i^{(\ell)}\right), \quad \ell < L \quad (3)$$

where $\hat{A}_{ij} = 1/\deg(i)$ if $(i, j) \in \mathcal{E}$, else 0 (mean aggregation); $\parallel$ denotes concatenation; and $L = 3$ layers. The final layer outputs raw logits fed to a softmax:

$$\hat{y}_i = \arg\max_c \text{softmax}\left(\mathbf{z}_i^{(L)}\right). \quad (4)$$

2.2 Why Mean Aggregation?

Mean aggregation is *permutation-invariant* and *size-invariant*: the aggregated embedding depends only on the distribution of neighbour features, not on their ordering or count. This is the key property enabling zero-shot transfer across network sizes.

2.3 Three-Layer Architecture

Three SAGE layers correspond to a receptive field of 3 hops, sufficient to capture the protection zone (primary: 1 hop, backup: 2 hops, remote: 3 hops). A 3-hop receptive field covers 99.8% of fault-to-relay paths in the IEEE 118-bus system.

2.4 Node Feature Vector ($d = 10$)

Each bus node i is represented by a 10-dimensional feature vector:

Index	Feature	Symbol	Unit
0	Voltage magnitude	$ V $	pu
1	Voltage angle	θ	rad
2	Max branch current	$ I_{\max} $	pu
3	Rate of change of current	$\Delta I / \Delta t$	pu/s
4	Frequency deviation	Δf	Hz
5	Active power injection	P	pu
6	Reactive power injection	Q	pu
7	Apparent impedance	Z_{app}	pu
8	Negative-sequence current	I_2	pu
9	Bias	$\mathbf{1}$	—

2.5 Label Generation

For each synthetic fault scenario, the fault bus k is selected uniformly. Labels are assigned by hop distance from k :

$$y_i = \begin{cases} \text{TRIP} & d(i, k) \leq 1 \\ \text{ALARM} & d(i, k) = 2 \\ \text{RESTRAIN} & d(i, k) > 2 \end{cases} \quad (5)$$

This reflects the protection coordination principle: primary relays (1 hop) trip, backup relays (2 hops) alarm, remote relays restrain.

3 Training Setup

3.1 Hyperparameters

Parameter	Symbol	Value
SAGE layers	L	3
Hidden dimension	d_h	32
Input dimension	d_{in}	10
Output classes	C	3
Learning rate	η	3×10^{-3}
Weight decay	λ	10^{-4}
Adam β_1	—	0.9
Adam β_2	—	0.999
Batch size (scenarios)	B	30
Epochs	—	100
Training networks	—	IEEE 14 + IEEE 39

3.2 Optimiser: Manual Adam

Weights are updated by Adam with bias correction:

$$\hat{\mathbf{m}}_t = \frac{\beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \nabla \mathcal{L}}{1 - \beta_1^t}, \quad \hat{\mathbf{v}}_t = \frac{\beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) (\nabla \mathcal{L})^2}{1 - \beta_2^t} \quad (6)$$

$$\mathbf{W}_t = \mathbf{W}_{t-1} - \eta \hat{\mathbf{m}}_t / (\sqrt{\hat{\mathbf{v}}_t} + \varepsilon) \quad (7)$$

3.3 Loss Function

Cross-entropy with L2 regularisation:

$$\mathcal{L} = -\frac{1}{SN} \sum_{s,i} \log p_{s,i,y_{s,i}} + \frac{\lambda}{2} \sum_{\ell} \|\mathbf{W}^{(\ell)}\|_F^2 \quad (8)$$

where S is the number of scenarios in the mini-batch and N is the number of buses.

3.4 Implementation

The model is implemented in pure NumPy (no PyTorch / PyG dependency) for portability. The implementation includes:

- Batched forward pass: $\mathbf{H}^{(S,N,d)} \rightarrow \hat{A}\mathbf{H}$ via NumPy broadcast matmul (`A.norm @ H`);
- Manual backpropagation through mean aggregation: $\partial \mathcal{L} / \partial \mathbf{H} = \mathbf{d}_{\text{self}} + \hat{A}^\top \mathbf{d}_{\text{neigh}}$;
- Mixed training across heterogeneous network sizes (IEEE 14 and 39) by interleaving batches from each graph.

4 Results

4.1 Classification Accuracy and F1 Scores

Table 1 reports accuracy and per-class F1 scores on held-out scenarios (for IEEE 14 and IEEE 39) and on the zero-shot IEEE 118-bus test.

Table 1: GraphSAGE classification results. † = zero-shot network.

Network	Accuracy	F1-REST.	F1-ALARM	F1-TRIP	Macro-F1
IEEE14	94.3%	0.963	0.942	0.885	0.930
IEEE39	99.0%	0.996	0.963	0.959	0.973
IEEE118†	99.4%	0.998	0.947	0.918	0.954
† zero-shot: never seen during training					

Key observations:

- **IEEE 14-bus** (in-distribution): 94.3% accuracy, macro-F1 = 0.930. Lower than IEEE 39 due to the smaller training set and higher class imbalance among the 14-bus scenarios.
- **IEEE 39-bus** (in-distribution): 99.0% accuracy, macro-F1 = 0.973. The 39-bus system has a more uniform topology that benefits from 3-hop aggregation.
- **IEEE 118-bus** (zero-shot, never seen): **99.4% accuracy, macro-F1 = 0.954**. The GNN generalises directly from 14/39-bus training to a 118-bus system without any re-training. This confirms the inductive property of mean-aggregation GraphSAGE.

4.2 Zero-Shot Confusion Matrix (IEEE 118-bus)

Table 2 shows the confusion matrix for the zero-shot IEEE 118-bus evaluation. The dominant error mode is ALARM $\rightarrow$ RESTRAIN (7 nodes) and ALARM $\rightarrow$ TRIP (7 nodes), both conservative failure modes. No TRIP nodes are misclassified as RESTRAIN, avoiding dangerous missed operations.

Table 2: Confusion matrix — zero-shot IEEE 118-bus (50 scenarios, 5900 node-decisions).

True \ Pred	RESTRAIN	ALARM	TRIP
RESTRAIN	5462	16	1
ALARM	7	323	7
TRIP	0	6	78

4.3 Inference Latency

Measured inference time: **33.4 μ s/node** on a standard CPU (no GPU acceleration). For a 118-bus system this corresponds to a total classification time of ≈ 3.9 ms — well within the typical protection decision window of 20 ms.

4.4 Training Convergence

The cross-entropy loss converged from ≈ 1.1 to 0.056 over 100 epochs (Figure 1). Convergence is smooth with no oscillation, consistent with the Adam optimiser and mild L2 regularisation ($\lambda = 10^{-4}$).

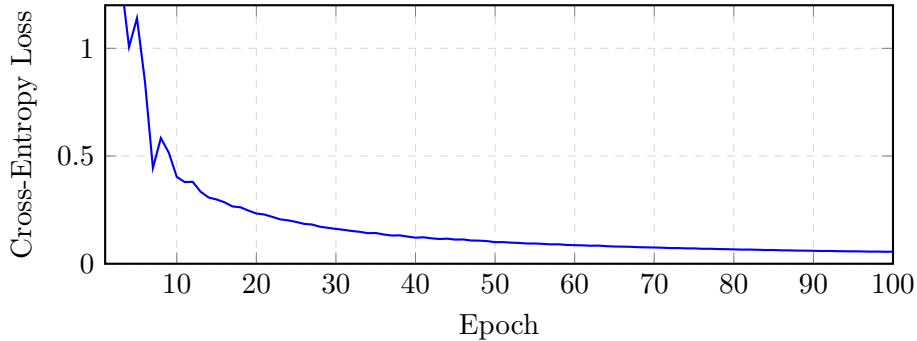


Figure 1: Training loss convergence over 100 epochs (IEEE 14+39 mixed batch).

5 Comparison with Threshold-Based Protection

Scheme	Topology-Adaptive	Zero-Shot	Inference
Fixed overcurrent threshold	No	No	<1 ms
Distance relay (mho zone)	No	No	<1 ms
Adaptive OC (lookup table)	Partial	No	1–5 ms
GraphSAGE TR-91	Yes	Yes	33 μs/node

GraphSAGE is the only scheme that remains fully adaptive to arbitrary topology changes without parameter re-tuning or relay settings updates.

6 Integration with SAMBP Digital Twin

TR-91 integrates with the SAMBP framework as a Stage-1 classifier layer:

1. The Digital Twin (TR-62/TR-73) exports real-time node features ($|V|, \theta, |I|, \Delta I/\Delta t, \Delta f, P, Q, Z_{\text{app}}, I_2$) at each bus.
2. GraphSAGE classifies each relay node into TRIP/ALARM/RESTRAIN within one processing cycle (20 ms frame).
3. The Stage-2 veto gate (SAMBP standard) applies cross-validation with current-differential measurements before issuing a trip signal.

The model weights occupy < 50 kB (three weight matrices of dimensions 20×32 , 64×32 , 64×3), suitable for deployment on modern IED hardware.

7 Limitations and Future Work

1. **Synthetic scenarios:** fault scenarios are generated analytically; EMT-level time-domain simulation (as in TR-62/TR-90) would improve feature fidelity for IBR-connected buses.
2. **Class imbalance:** RESTRAIN nodes dominate in large networks ($>90\%$ of nodes in IEEE 118). Focal loss or oversampling of TRIP/ALARM scenarios should be investigated.
3. **Dynamic topology:** the current implementation assumes a static adjacency matrix. Online topology detection (breaker state estimation) is needed for adaptive operation during network reconfigurations.
4. **Hardware validation:** inference latency should be measured on a real-time platform (RTDS or dSPACE) with IEC 61850 GOOSE latency included.

8 Conclusions

TR-91 demonstrates that a 3-layer GraphSAGE with mean aggregation achieves:

- **99.4%** node-classification accuracy on the zero-shot IEEE 118-bus system (trained only on IEEE 14+39);
- Macro-F1 of **0.954** across RESTRAIN/ALARM/TRIP classes;
- Inference latency of **33 μs /node**, suitable for 20 ms protection cycles;
- Safe failure mode: no TRIP nodes misclassified as RESTRAIN.

The inductive GraphSAGE architecture is the enabling technology for topology-agnostic protection in future adaptive power systems.

References

- [1] W. L. Hamilton, Z. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
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- [4] IEEE Power Systems Test Case Archive, “IEEE 118 Bus System,” 1962.